Eric R. Van Clepper

Contact Information

Wisconsin Center for Origins Research (WiCOR)
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ORCID

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Education

2020 - 2026	Ph.D. in Geophysical Sciences, University of Chicago, Chicago, IL *Thesis Topic: Coupled Chemistry and Dynamics in Protoplanetary Disks* *Advisor: Fred Ciesla*
2015 - 2019	BA *cum laude* Astronomy & Planetary Science with Honors, Northwestern University, Evanston, IL *Advisor: Giles Novak*

Employment

2026 -	WiCOR Postdoctoral Fellow, University of Wisconsin-Madison, Madison, WI
2020 - 2026	Research and Teaching Assistant, University of Chicago, Chicago, IL
2019 - 2020	N'Cat Tutor, Northwestern University, Evanston, IL
2017 - 2019	Undergraduate Research Assistant, Northwestern University, Evanston, IL

Selected Grants

Oct 2023 - Oct 2026	NASA FINESST
Sept 2020 - Sept 2022	McCormick Fellowship
June 2018 - Sept 2018	Northwestern Weinberg Research Grant
June 2018 - Sept 2018	NASA ISG Summer Research Grant
June 2017 - Sept 2017	NASA ISG Summer Research Grant

Conference Presentations

August 2024	Goldschmidt Conference, Chicago
January 2024	FARGO3D 1st Workshop, Santiago, Chile
August 2023	Meteoritical Society, 86th Annual Meeting, Los Angeles
July 2023	Kavli-IAU Astrochemistry Symposium, Astrochemistry VIII, Traverse City
August 2021	Meteoritical Society, 84th Annual Meeting, Chicago

Poster Presentations

September 2026	Origins of Life, Origin Federation Conference
October 2025	Solar System in Context, NOIRLab
September 2025	Origins of Life, Origin Federation Conference
June 2025	Origins of Solar Systems, Gordon Research Conference
July 2024	New Heights in Planet Formation, ESO Dust Busters
June 2023	Origins of Solar Systems, Gordon Research Conference
August 2022	Emerging Researchers in Exoplanet Sciences VII

Software Development

`CANDY` https://github.com/ervc/newcandy

Chemistry ANd DYnamics, coupled astrochemistry and dust evolution for self-consistent modeling of chemical composition during planetesimal formation.

`ParTrace` https://github.com/ervc/cpartrace

Particle Tracking algorithm to track trajectories of tracer particles in 3D Hydrodynamic disk simulations, accounting for gas drag with 3D flows around embedded planets.

Professional Activities

- Executive Officer for NASA Review Panel.
- Peer Reviewer for *The Astrophysical Journal*.

Departmental Service

2024	Lead TA for intro to coding Bootcamp for incoming students
2024 - 2025	Graduate Union Steward for the Department of Geophysical Sciences
2023	Co-organizer of University of Chicago Graduate Research Expo
2022 - 2023	Co-organizer of University of Chicago Exoplanet Journal Club

Outreach

Nov 18, 2025	Meet a Scientist grad student life panel, College of the South
July 23, 2025	Guest Lecture on planet formation for UChicago summer class ASTR11901
Nov 14, 2023	Planetary science and grad student life panel, College of the South
Nov 21, 2022	Planetary science and grad student life panel, College of the South

Courses Taught

As TA at University of Chicago

Winter 2025	PHSC 13600	Natural Hazards
Spring 2024	GEOS 21200	Physics of the Earth
Autumn 2021	PHSC 10800	Earth as a Planet: Exploring our Place in the Universe
Winter 2021	PHSC 10100	Origin and Evolution of the Solar System and the Earth

Publications

First Author Publications

Van Clepper, E. R., E. Bergin, F. Alarcón, and F. Ciesla. 2025. "Dust Recycling and Icy Volatile Enhancement (DRIVE): A Novel Method of Volatile Enrichment in Cold Giant Planets." *The Astrophysical Journal Letters* 994: L44. https://doi.org/10.3847/2041-8213/ae1d81

Van Clepper, E., E. Price, and F. J. Ciesla. 2025. "Three-Dimensional Transport of Solids in a Protoplanetary Disk Containing a Growing Giant Planet." *The Astrophysical Journal* 980: 201. https://doi.org/10.3847/1538-4357/ada8a4.

Van Clepper, E., J. B. Bergner, A. D. Bosman, E. Bergin, and F. J. Ciesla. 2022. "Chemical Feedback of Pebble Growth: Impacts on CO Depletion and C/O Ratios." *The Astrophysical Journal* 927 (2): 206. https://doi.org/10.3847/1538-4357/ac511b.

Other Publications

Ciesla, F. J., E. Van Clepper, J. Bergner, and E. Bergin. 2025. "Oxygen Isotope Constraints on the Importance of Photochemical Processing in Protoplanetary Disks." *The Astronomical Journal, In Press*. https://arxiv.org/pdf/2512.04944

Price, E. M., E. Van Clepper, and F. J. Ciesla. 2025. "Dynamics of Small, Constant-Size Particles in a Protoplanetary Disk with an Embedded Protoplanet." *The Astrophysical Journal* 979 (1): 37. https://doi.org/10.3847/1538-4357/ad9f35.

Seligman, D. Z., L. A. Rogers, S. H. C. Cabot, et al. incl. E. Van Clepper. 2022. "The Volatile Carbon-to-Oxygen Ratio as a Tracer for the Formation Locations of Interstellar Comets." *The Planetary Science Journal* 3 (7): 150. https://doi.org/10.3847/PSJ/ac75b5.

Santos, F. P., D. T. Chuss, C. D. Dowell, et al. incl. E. Van Camp.* 2019. "The Far-Infrared Polarization Spectrum of ρ Ophiuchi A from HAWC+/SOFIA Observations." *The Astrophysical Journal* 882 (2): 113. https://doi.org/10.3847/1538-4357/ab3407.

Chuss, D. T., B-G Andersson, J. Bally, et al. incl. E. Van Camp.* 2019. "HAWC+/SOFIA Multiwavelength Polarimetric Observations of OMC-1." The Astrophysical Journal 872 (2): 187. https://doi.org/10.3847/1538-4357/aafd37.

**Name Change from Eric Van Camp to Eric Van Clepper in 2021*